MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
COURSE CODE- MG4DSCFTQ200
COURSE TITLE: FOOD CHEMISTRY

Duration: 1.5 hrs Maximum Marks: 50

Part A
Short Answer Type Questions
Answer any 5 questions
Each question carries 2 Marks

1. What do you mean by the term birefrigence? [R] [CO 2, 3, 5]
2. Define dextrose equivalence. Give its significance. [An] [CO 2, 3, 5]
3. Differentiate between thermoset and thermoplastic gels. [U] [CO 2, 3, 5]
4. Apples turn brown when cut-Give reasons to substantiate the statement. [R] [CO 1, 2]
5. What is instant starch? Give its applications. [U] [CO 2, 3, 5]
6. What is hydrogenation? State its importance. [A] [CO 4]
7. What is flavour reversion? [U] [CO 1,2]

(5x2=10 marks)

Part B
Short Essay Type Questions
Answer any 5 questions
Each question carries 4 Marks

8. Describe the mechanism of racemisation of proteins in food during processing. [R] [CO 2, 3, 5]
9. Narrate the major events during staling of bread. [R] [CO 2, 3, 5]
10. Evaluate the physical and chemical changes during frying of oils. [U] [CO 1, 2]
11. Write a note on the types of immobilisation of enzymes. [U] [CO 1, 2]
12. Elucidate the physical and chemical properties of fats. [E] [CO 1, 2]
13. Give a brief account on the modification of cellulose in food. [U] [CO 2, 3, 5]
14. Discuss the different cross-linked and substituted starch used in foods. [U] [CO 1, 2]

(5x4=20 marks)

Part C
Long Essay Type Questions

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Answer any **2** questions
Each question carries **10** Marks

15. Explain in detail the process of gelatinisation and retrogradation of starch. [U] [CO 2, 3, 5]
16. Give a detailed account of the chemical reactions involved in Maillard reaction. [U] [CO 1, 2]
17. Describe in detail the major process involved in the refining of fats and oils [A] [CO 4]
18. Elaborate on the functional properties of proteins in food. [U] [CO 2, 3, 5]

(2x10=20 marks)

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
COURSE CODE: MG4DSCFTQ201
COURSE TITLE: FOOD MICROBIOLOGY

Duration: 1.30 hrs Maximum Marks: 50

Part A

Answer in one word
Answer any 5 questions
Each question carries **2** Mark

1. Define bacterial capsule. [R] [1]
2. What is the function of a phase contrast microscope?? [U] [1]
3. Differentiate between selective media and differential media. [R] [1]
4. Define aseptic technique [U] [4]
5. Define diauxic growth [R] [2]
6. What is food intoxication? Give one example. [A] [3]
7. List any two factors affecting microbial growth in food [E] [2]

(5x2=10 marks)

Part B

Short Essay Type Questions
Answer any **5** questions
Each question carries **4** Marks

8. Explain the contributions of Louis Pasteur to microbiology. [R] [1]
9. Describe the principle and applications of electron microscopy [U] [1]
10. Explain the structure of Gram-positive and Gram-negative bacterial cell walls [An] [2]
11. Discuss the bacterial growth curve with a neat diagram. [U] [2]
12. Differentiate between food infection and food intoxication with examples. [An]

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[3]

13. Describe the role of selective and differential media in identifying pathogens [An] [4]

14. Describe on the different sterilization techniques [U] [2]

(5x4=20 marks)

Part C

Long Essay Type Questions
Answer any **2** questions
Each question carries **10** Marks

15.Describe the structure and functions of bacterial cell components (cell wall, membrane, flagella, pili, capsule, endospore). [An] [2]

16. Explain the methods of sterilization and their applications in food microbiology laboratories and industries. [A] [4]

17. Discuss major foodborne diseases caused by bacteria, their causative agents, symptoms, and preventive measures [R] [2]

18. Explain the principles and applications of different microscopy techniques used in microbiology (bright field, dark field, phase contrast, fluorescence, TEM, SEM). [U] [1]

(2x10=20 marks)

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

COURSE CODE: MG4DSCFTQ202

COURSE TITLE: NOVEL TECHNOLOGIES IN FOOD

Duration:1.5hrs Maximum Marks:50

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. State the significance of ethylene scavengers in packages. [R] [CO3]
2. Draw a diagrammatic representation of aseptic packaging process. [R] [CO3]
3. List out some examples of preservative releasers applied in active packaging. [R] [CO3]
4. Outline the properties of Near Critical Fluids. [R] [CO2]
5. List the advantages of IR drying. [R] [CO2]
6. Write down the principle behind PEF. [A] [CO2]
7. Describe pulsed light technology. [U] [CO4]

(5×2 = 10 marks)

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

8. Write a short note on Active packaging. [A] [CO3]

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9. Write a note on indicators used in intelligent packaging. [A] [CO3]

10. Explain the various applications of SCFE in food processing. [U] [CO2]

11. Write a short note on microwave blanching. [A] [CO2]

12. Discuss the applications of ultra-sonication in food processing. [U] [CO4]

13. Explain the mechanism of Ohmic heating. [U] [CO4]

14. Describe inductive heating. [U] [CO4]

(5×4 = 20 marks)

Part C

Long Essay Type Questions

Answer any 2 questions

Each question carries10 marks

15. Explain Active packaging. What is the importance of such packaging systems for high quality products? [U] [CO3]

16. Explain in detail: - a) HPP b) PEF c) Microwave drying [U] [CO2]

17. Explain in detail the principle of cold plasma technology, its generation systems and its application in foods. [U] [CO2]

18. Explain in detail: - a) High intensity light technology b) Inductive cooking [U] [CO4]

(2×10 = 20 marks)

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MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
COURSE CODE: MG4DSEFTQ200
COURSE TITLE: FOOD PACKAGING TECHNOLOGY

Duration: 2 hrs Maximum Marks: 70

Part A
Short answer
Answer any 10 questions
Each question carries 2 Marks

1. What is the primary function of food packaging? [U] [1]

2 Give two examples for rigid packaging.[A] [4]

3. Define Aseptic packaging. [R] [5]

4. What is meant by a biodegradable package? [A] [5]

5. Expand the terms MAP and CAP.[R] [5]

6. Name any one packaging regulation that governs food labelling and packaging in India. [U] [2]

7. Define WTR & GTR. [R] [5]

8. List any four advantages of glass, as a food packing material. [An] [3]

9. Name any one physical or chemical test performed on packaging material [U] [5]

10. State the importance of tin plate in food packaging [U] [3]

11. What is meant by paperboard? Give two examples of paper board used in food packaging.[R] [1]

12. Write about lacquers used in food cans. [U] [5]

Part B
Short essay type questions
Answer any 6 questions
Each question carries 5 Marks

13. Discuss glass as a food packaging material with its properties, uses & demerits. [R] [1]

14. What are retort pouches? Describe its advantages over metal cans in food packaging. [A] [4]


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